Varun Karanda

Frontend Engineer – II

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PROFESSIONAL SUMMARY

Frontend Engineer with 5 years of experience building scalable web applications using React.js, TypeScript, and modern JavaScript (ES6+). Skilled in state management (Redux, Redux-Saga), RESTful API integration, and performance optimization. Core contributor to **K-SMART**, a state-scale platform supporting **35M+** users. Experienced in complex form workflows, asynchronous data handling, and production debugging. Exploring AI applications using LangChain.

TECHNICAL SKILLS

Frontend	React.js, JavaScript, TypeScript, React Hooks, Redux Toolkit, Redux-Saga, React Hook Form
UI & Styling	Tailwind CSS, Chakra UI, Styled Components, HTML5, CSS3
Backend & APIs	Python (Flask, Django), Node.js, SQL, REST APIs, API Integration
Tools	Git, GitHub, Redis, AWS S3, Puppeteer, K9s
Testing	Vitest, Unit Testing, Component Testing, Test-Driven Debugging
AI & Emerging	LangChain, LLM Concepts, Agentic AI
Practices	Agile, Code Review, Responsive Design, Accessibility, Debugging, Problem Solving

PROFESSIONAL EXPERIENCE

Edstem Technologies Pvt. Ltd.

Kochi, Kerala, India

Software Engineer 2

May 2024 – Present

- Led end-to-end frontend delivery for **K-SMART's** Hindu Marriage (Civil Registration) and Rent Module (Property Tax), contributing to design decisions for scalable multi-step workflows.
- Drove integration and enhancement of civil registration workflows **(Birth, Death & Marriage)**, supporting **35M+** users.
- Contributed to the integration of **150+ REST APIs** across large-scale service modules, implementing centralized error handling and diagnosing production issues using **K9s** and **Kubernetes logs.**
- Partnered on building a document generation service using **Puppeteer, Redis, and AWS S3**, contributing to **Redis-based job** deduplication and **S3 caching** to reduce certificate retrieval time from multi-second rendering to sub-second responses under peak traffic.
- Led and mentored a frontend **team of 8**, conducting structured code reviews with **24–48** hour turnaround, and collaborating with backend and QA teams to maintain release quality.

Software Engineer

Jun 2022 – Apr 2024

- Developed and maintained the shared **React component library** used across **K-SMART** modules, establishing reusable UI patterns that improved consistency and reduced duplication across the platform.
- Initiated development of core civil registration workflows **(Birth, Death & Marriage)**, establishing the foundation for multi-step forms with dynamic validation and role-based access.
- Implemented complex registration flows using **React Hook Form** and **Redux-Saga**, improving data consistency and reducing validation errors by **~20%** across citizen-facing modules.
- Integrated REST APIs across multiple service modules, managing asynchronous state using **Redux-Saga** and contributing to centralized error handling patterns across the platform.

Junior Software Engineer

Jun 2021 – May 2022

- Built a feature-rich expert marketplace (RecSperts – USA) using React, implementing multi-criteria search, real-time booking workflows, and provider analytics dashboards.
- Built compliance and risk visualisation dashboards for Crown Jewel – an Australian enterprise cybersecurity platform – translating complex asset-protection data into interactive panels for non-technical stakeholders.

My Softnet

Kerala, India

Frontend Intern

Nov 2020 – May 2021

- Developed responsive and accessible UI components using React.js, and contributed to identifying and resolving API integration issues in a production environment.
- Gained hands-on experience across the frontend development lifecycle, including translating design mockups into UI and contributing to feature releases in a collaborative development environment.

KEY PROJECTS

K-SMART Civil Registration Platform — *Government of Kerala*

State-wide e-governance platform unifying 15+ modules for Local Self Government Institutions, covering civil registration, property tax, building permits, pensions, finance, and more, with real-time tracking and digital certificate issuance, supporting **35M+ users**.

PDF & Certificate Generation Service — *Government of Kerala*

Asynchronous document generation pipeline using Puppeteer, Redis, and AWS S3, designed for high-volume certificate processing, optimizing performance through caching and request deduplication.

RecSperts – Expert Marketplace — *USA*

Marketplace platform connecting service providers with users across multiple categories, featuring authentication, booking workflows, reviews, and analytics dashboards, with reusable UI components for scalability and a consistent, responsive design.

Crown Jewel – Cyber Resilience System — *Australia*

Enterprise cybersecurity platform providing compliance and risk visualization through interactive dashboards, translating complex asset-protection data into actionable insights for non-technical stakeholders, with real-time monitoring and exportable reports.

CERTIFICATIONS

- Microsoft Azure AI Fundamentals (AI-900) [Credential Available]
- Claude Code in Action — Anthropic [Credential Available]
- Gemini CLI: Code & Create with an Open-Source Agent — Google
- Agent Skills with Anthropic — Anthropic
- Agentic AI — DeepLearning.AI [Credential Available]
- LangChain for LLM Application Development — DeepLearning.AI / Andrew Ng

EDUCATION

Bachelor of Technology (B.Tech)

2010 – 2014

College of Engineering Trikaripur, Kerala, India